Automation is a core tenet of cloud native. The idea is to automate repetitive manual tasks to accelerate the delivery and deployment of cloud native applications. Many tasks can be automated, from building applications to deploying them, from provisioning infrastructure to managing configuration. The most important advantage of automation is that it makes processes and tasks repeatable and overall systems more stable and reliable. Manually executing a task is error-prone and costs money. By automating it, we can get a result that is both more reliable and more efficient.

In the cloud computing model, computing resources are provisioned in an automated, self-service model, and they can be increased or decreased elastically. Two significant categories of automation for the cloud are infrastructure provisioning and configuration management. We call them *infrastructure as code* and *configuration as code*.

Martin Fowler defines *infrastructure as code* as "the approach to defining computing and network infrastructure through source code that can then be treated just like any software system."[10]

Cloud providers offer convenient APIs for creating and provisioning servers, networks, and storage. By automating those tasks with a tool like Terraform, putting the code in source control, and applying the same test and delivery practices used for application development, we get a more reliable and predictable infrastructure, which is reproducible, more efficient, and less risky. A simple example of an automated task could be creating a new virtual machine with 8 CPUs, 64 GB of memory, and Ubuntu 22.04 as the operating system.

After we've provisioned computing resources, we can manage them and automate their configuration. Paraphrasing the previous definition, *configuration as code* is the approach to defining the configuration of